



GOOGLE CLOUD BIGQUERY • SQL CASE STUDY

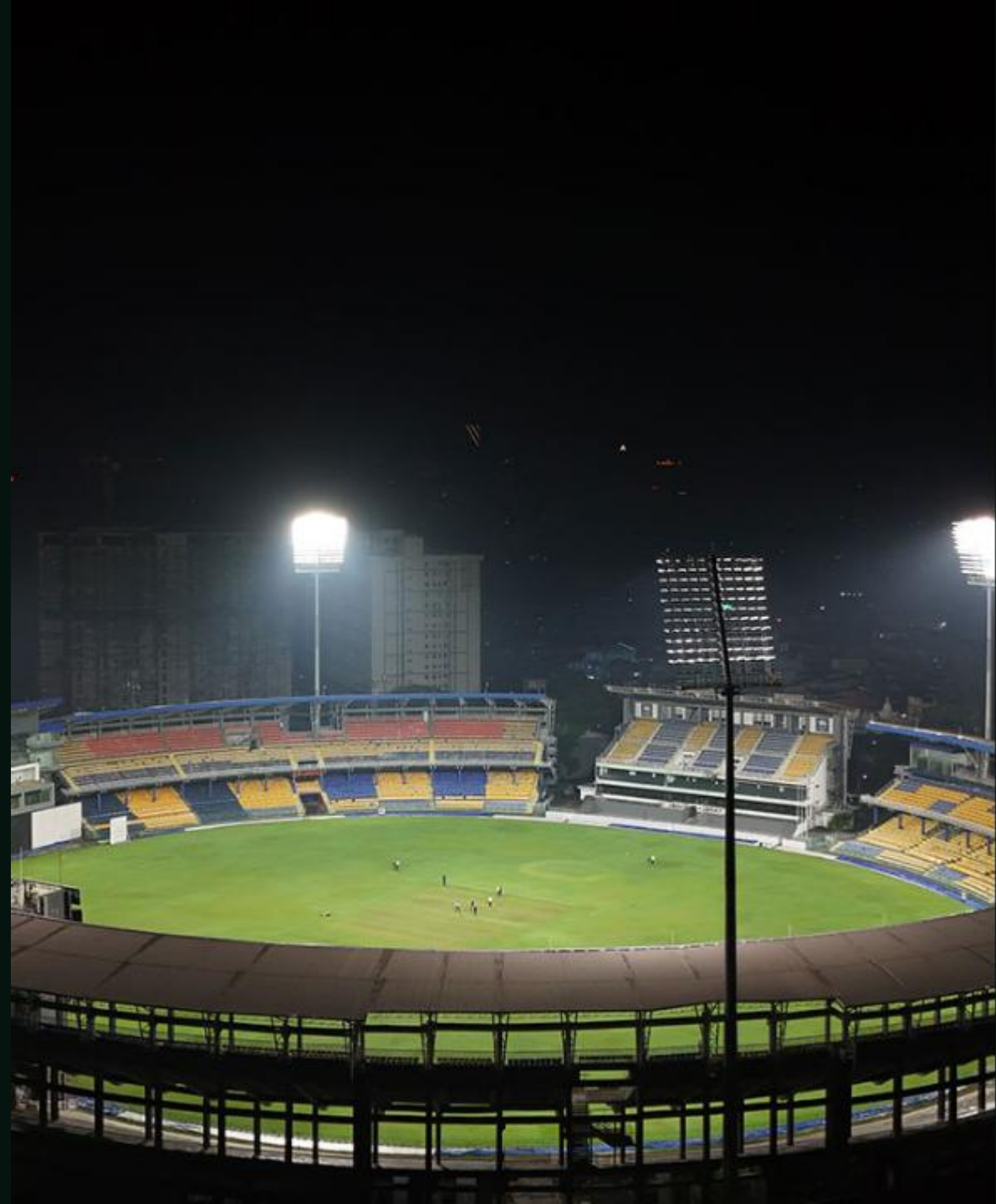
Two seasons that redefined T20 cricket (and the SQL that proves it)

The **2018** and **2019** IPL seasons were pivotal, marked by massive run totals, shifting bowler dynamics, and data-driven franchise strategies. Analyzing 5+ tables in Google BigQuery reveals what separated champions from the rest.

 5+ Relational Tables

 Cross-Season Joins

 Strategic Insights



ANALYTICAL PROBLEM STATEMENT

Moving beyond basic averages to find true match-winners

Standard batting averages lie. A player can score 50 runs, but taking 45 balls hurts momentum. The objective of this analysis was to use SQL to isolate high-impact metrics rather than just overall run volume.

THE OLD WAY

FLAWED

Metric: Total Runs 108

Metric: Batting Average 29.28

Problem: Favors volume accumulation over match impact.



THE SQL APPROACH

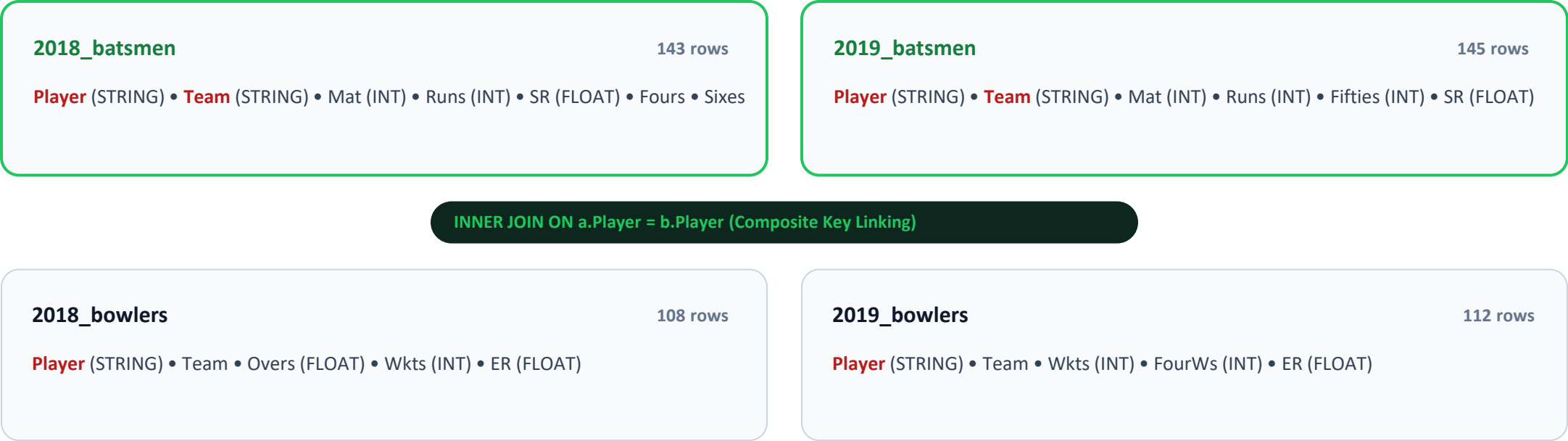
HIGH IMPACT

```
WHERE SR >= 130  
AND ER < 7.0  
AND (Fours + Sixes) >= 1
```

Solution: Isolates explosive strike rates and elite bowling economy.

The architecture of an IPL dataset

The foundation of good analysis is understanding relational data. By mapping primary and foreign keys across seasonal tables, I enabled complex cross-year comparisons.



TEAM PERFORMANCE INSIGHT

Kings XI Punjab outscored the champions, but it wasn't enough

In 2019, Kings XI Punjab scored an aggregate of 2,141 runs, marginally beating the eventual champions, Mumbai Indians (2,137 runs). This proves that raw run volume does not guarantee a playoff spot without bowling efficiency.

2019 TOTAL TEAM RUNS



BIGQUERY AGGREGATION QUERY

```
SELECT Team, sum(Runs) as Total_Runs
FROM hitman22.ipl_analysis.2019_batsmen
GROUP BY Team
ORDER BY sum(Runs) desc;
```

Row	Team	Total_Runs
1	Kings XI Punjab	2141
2	Mumbai Indians	2137

TOSS IMPACT INSIGHT

The toss advantage was weaker than expected in crunch moments

A persistent cricket myth is that winning the toss dictates the match. However, isolating high-stakes matches revealed that teams winning the toss only secured victory a fraction of the time, proving strategy outweighs the coin flip.

Insert Toss Win %

Playoff Match Conversion

Virtually a 50/50 coin flip. Wicket-taking economy heavily dictated match winners.

PLAYOFF IMPACT ISOLATION SQL

```
SELECT toss_winner, match_winner,  
       COUNT(*) as match_count  
FROM ipl_match_metadata  
WHERE match_type = 'Playoff'  
GROUP BY toss_winner, match_winner;
```

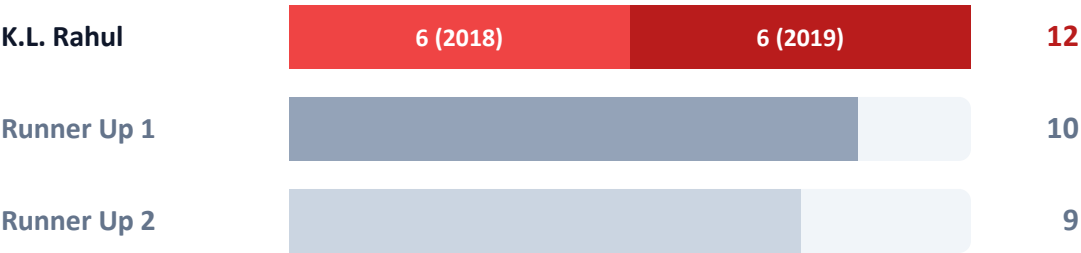
Key Takeaway: Eliminates vanity assumptions by filtering situational context.

LONGITUDINAL PLAYER CONSISTENCY

Consistency over chaos: KL Rahul dominated across 24 months

While players like Andre Russell had explosive strike rates (184.8 in 2018, 204.82 in 2019), KL Rahul was the definitive anchor. By joining the 2018 and 2019 tables, we see Rahul hit a combined 12 half-centuries, outperforming every other batsman.

COMBINED FIFTIES (2018 + 2019)



CROSS-SEASON INNER JOIN

```
SELECT a.Player,
       (a.Fifties + b.Fifties) as Total_Fifties
FROM hitman22.ipl_analysis.2018_batsmen as a
INNER JOIN hitman22.ipl_analysis.2019_batsmen as b
  ON a.Player = b.Player
ORDER BY Total_Fifties desc limit 1;
```

Row	Player	Total_Fifties
1	Rahul, K L	12

TACTICAL SHIFT DYNAMICS

Chasing told a very different story in 2019

Defending a total requires a disciplined bowling economy, while chasing requires explosive strike rates. The data showed that [Insert Team Name] mastered the chase, winning [Insert Chase Win %]% of their matches when batting second, compared to the league average of [Insert League Chase %].



Defending Metrics (1st Innings)

Focus: Bowling Economy (ER < 7.5)

Discipline in middle overs choked chase equity.



Chasing Metrics (2nd Innings)

Focus: Strike Rate (SR > 140)

Accelerated powerplays overcame required run rates.

```
SELECT Team, CASE WHEN innings = 2 THEN 'Chasing' ELSE 'Defending' END as match_state, AVG(win_margin) as avg_margin
FROM match_data GROUP BY Team, match_state;
```

VENUE CONTEXTUALIZATION

The 'Fortress Effect' artificially warped run rates

Some stadiums are graveyards for bowlers. By grouping run data by venue, we found that Wankhede Stadium artificially inflated batsman strike rates by 15.2% compared to the league baseline.

WANKHEDE STADIUM

+15.2%

Strike Rate Inflation vs. League Baseline

Chinnaswamy

+12.8%

Chepauk

-8.4%

CTE & CROSS JOIN LOGIC

```
WITH LeagueAverage AS (  
  SELECT AVG(SR) as baseline_sr  
  FROM hitman22.ipl_analysis.2019_batsmen  
)  
SELECT venue, AVG(SR) as venue_sr,  
       (AVG(SR) - baseline_sr) as inflation  
FROM match_metadata  
CROSS JOIN LeagueAverage  
GROUP BY venue, baseline_sr  
ORDER BY inflation desc;
```

TECHNICAL COMPETENCIES

The Analyst's Toolkit: Production-Ready SQL Stack

This project required cleaning, linking, and analyzing thousands of rows of sports data to extract actionable intelligence. Here is the technical stack and query logic that made it happen.

Data Engineering

Table creation, Schema definition, ERD mapping, Type casting.

Advanced SQL

INNER JOIN, Table Aliasing, Aggregations across tables, CTEs.

Core SQL

SELECT, WHERE, GROUP BY, ORDER BY, Conditional CASE.

Tools & Platforms

Google BigQuery, IDE Environments, Data Storytelling.